

Grade Eight Science - Static Electricity Review Questions

MYP Command Terms that are used on this problem set:

State: Give a specific name, value, or other brief answer without explanation of calculator	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Recall: Remember or recognize from prior learning experience	Construct: Display information in a diagrammatic or logical form
Analyze: To identify parts and relationships, and to interpret information to reach conclusions	Interpret: Use knowledge and understanding to recognize trends and draw conclusions from given information

1. **Apply** scientific knowledge to **state** the charge on each of the following two items when they are rubbed together (Level 3-4)

Charging situation	Charge on item 1	Charge on item 2
Polyester (1) rubbed with wool (2)	Negative	Positive
Glass rod (1) rubbed with silk (2)	Positive	Negative

2. **State** the type of charging happening in question 1 above and **state one way** that this type of charging compares to any other type of static charging studied in class (Level 3-4)

• The type of charging is charging by friction since the objects are rubbed together • Charging by friction compares to conduction because there is contact used in both types of charging
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3. A plastic rod is rubbed with a piece of silk and then the rod is touched to the ball of a neutral electroscope.

A. **State** the two types of static charging happening here, and **outline** your reasoning below (Level 3-4, Level 5-6)

- When the plastic rod is rubbed with silk this is charging by friction. I know this because friction means that two objects are rubbed together and the plastic rod is rubbed with the silk
- The second type of charging shown here is charging by conduction. Conduction means touch contact between objects and this occurs when the rod is touched to the ball of a neutral electroscope

B. When the plastic rod is rubbed with silk, **state** the charge left on the rod and the silk and **outline** your reasoning (Level 3-4, Level 5-6)

Since silk is below plastic on the triboelectric series it has a lower electron affinity than plastic. This means that when they are rubbed together the silk will put electrons onto the plastic making the silk positive because it has lost electrons and the plastic negative because it has gained electrons.

C. A student working in the lab would like to give a neutral electroscope a positive charge. **Apply** scientific knowledge to decide if the student should use the plastic rod or the silk from question "B," above? **Outline** your reasoning below (Level 5-6)

In order to make a neutral electroscope positive the student should use the silk from question B because the silk was positive as it gave electrons to the plastic.

By placing a positive piece of silk in contact with the ball of a neutral electroscope the silk will take negatives from the electroscope in order to neutralize itself. This will leave the electroscope with fewer negative charges than positive charges giving it an overall positive charge.

4. **Apply** information to make a scientifically supported judgement to indicate whether unknown "substance X," falls above or below the material given (Level 3-4)

Charging Situation	Is "substance X" below or above the given material on the triboelectric series
Rubbing cotton and "X" together gives "X" a positive charge	X is below cotton
Rubbing wool and "X" together gives "X" a negative charge	X is above wool

5. **State** any two examples of how you can achieve repulsion between two bar magnets (Level 3-4)

Repulsion can be shown between two bar magnets by putting like poles together. If you put the South Pole of one magnet near the South Pole of the other magnet, or the North Pole of one magnet near the North Pole of another magnet then repulsion will be shown as the magnets will move away from one another

6. A person wraps their hand around a negative styrofoam ball. **Outline** the purpose of doing this and **state** the charges that will be left on the hand and the ball. **Outline** your reasoning using both words and a diagram to support your answer below (Level 3-4, Level 5-6)

Words	Diagram
- The purpose of wrapping a neutral hand around a negative styrofoam ball is to ground the ball. This means to remove the charge from the ball bringing it back to neutral - When a neutral hand is wrapped around the negative ball the ball sees it as an opportunity to get itself back to neutral, so it gives negatives to the hand in order to even out its number of positives and negatives - At the end the ball will be neutral because it gave negatives to the hand and the hand will be negative because it took negatives from the	The diagram illustrates the process of grounding a negative styrofoam ball with a neutral hand. It is divided into two parts: 'During' and 'After'. In the 'During' part, a hand with '+' signs is shown wrapping around a ball with '-' signs. In the 'After' part, the ball is now neutral (labeled '(N)') and the hand is now negative (labeled '(-)' and has '-' signs).

ball	
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7. **Recall** that the type of subatomic particle that can move between objects is:

(Level 1-2)

- A. Protons
- B. Electrons and protons
- C. Neutrons
- D. Electrons
- E. Electrons and neutrons
- F. Electrons, protons, and neutrons

8. **Analyse** the following information and answer the question in the box below:

An ebonite rod is rubbed with paper. The paper is then rubbed with a neutral piece of polyester. The polyester is then rubbed with a neutral piece of fur. The fur is then brought near the ball of a neutral electroscope. As the fur is held near, a neutral plastic rod is touched to the aluminum ball of the electroscope. **Apply** your scientific knowledge about static charging to **describe** your reasoning to answer each of the following problems below (Level 7-8)

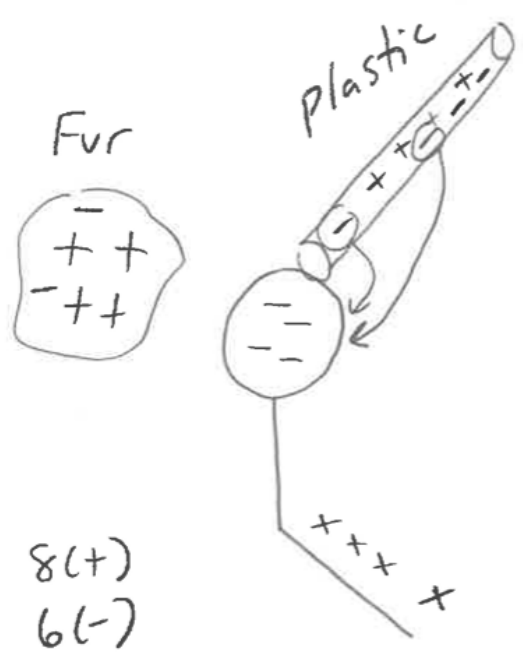
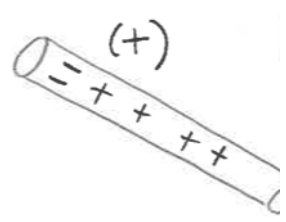
A. **Describe** how the ebonite rod, paper, polyester, fur, and electroscope got their charges. Your **description** must clearly indicate the charge left on each object and how each charge was obtained. Scientific language must be used in the **description**.

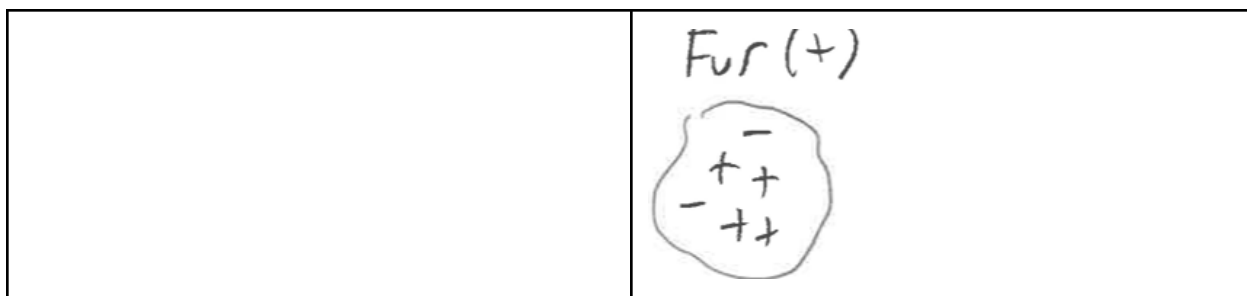
- Since paper has a lower electron affinity than ebonite when they are rubbed together the paper will put negatives on the ebonite making the ebonite Rod negative and the paper positive [Paper (+); Ebonite rod (-)]
- Since the paper is positive and the polyester is neutral, when they come into contact with one another the paper sees it as an opportunity to get itself back to neutral. The paper takes electrons from the polyester in order to neutralize itself. This makes the polyester positive. [polyester (+); paper (N)]
- Since the polyester is positive and the fur is neutral, when they come into contact with one another the polyester sees it as a chance to neutralize itself. The polyester takes negatives from the fur causing the fur to become positively charged and the polyester to become neutral. [polyester (N); fur (+)]
- When the positive fur is brought near the ball of a neutral electroscope, the charges on the electroscope will polarize. The negatives will come to the ball

because they are attracted to the fur. The positives will go to the leaf causing it to extend outward due to repulsion from the fur.

- The induction system (the fur and the electroscope) has a positive charge since the system has more positives than negatives
- When the neutral plastic rod is touched to the aluminum ball of the electroscope, the system sees it as a chance to get back to neutral. The plastic rod gives negatives to the electroscope to try and neutralize the positive system: [Electroscope (-); plastic rod (+)]

B. Using the description in question "A" above, **construct** each of the following diagrams

The induction between the fur and the neutral electroscope with the neutral plastic rod added (Charge transfer must be shown if there is any)	The fur, the plastic rod, and the electroscope with their final charges
 8(+) 6(-)	Electroscope: 
	Plastic Rod:  Fur:



9. When two magnets show attraction, **recall** that the behavior of the field lines is such that they clash and compress (Level 1-2)

A. True
B. False

10. A student working in the lab discovers that a small piece of iron is attracted to a horseshoe magnet, while a small piece of aluminum is not. **Outline** the reason why the iron shows attraction, but the aluminum does not. (Level 3-4)

The iron shows attraction while the aluminum does not because the domains of electrons in the piece of iron are aligned in the same direction as one another, while the domains of electrons in the aluminum are misaligned. Alignment of electrons is necessary in order to observe magnetism.

11. A neutral ebonite rod is touched to the ball of a positively charged electroscope. Once the rod and electroscope are separated the rod repels a ping pong ball. The ping pong ball then attracts some rice krispies which come near the ping pong ball, but don't touch it.

Apply scientific knowledge to figure out the final charge on the rod, the electroscope, the ping pong ball, and the rice krispies. **Describe** your reasoning below (Level 7-8)

Ebonite Rod	Electroscope	Ping Pong Ball	Rice Krispies
The ebonite rod will be positive because when it is touched to the ball of the positive electroscope, the electroscope saw it	The electroscope is neutral. When the electroscope was in contact with the ebonite rod it took electrons from the rod in order to	Since the ebonite rod is positively charged and the ping-pong ball repels the rod the ping-pong ball must also be positive.	Since the Rice Krispies were attracted to the positive ping pong ball. They have to be either negative or neutral.

as a chance to get itself back to neutral. The electroscope took negatives from the ebonite rod in order to make itself neutral and this left the rod positive	neutralize itself.	There is no neutrality in repulsion so the charges of the two objects must be the same in order for repulsion to be observed	Negative objects are attracted to positive objects. Neutral objects can be attracted to any charged objects because they can polarize resulting in attraction. There is not enough information given about the Rice Krispies to determine if they are negative or neutral, but we know they must be one or the other based on the reasoning above
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12. The Van De Graff generator used in class has a silicon roller at the bottom, a nylon roller at the top and a rubber belt that moves around both rollers. There is also a metal brush attached to the dome that touches the belt at the top.
Interpret the information given to answer the following questions about the generator. **Describe** your reasoning in the space provided (Level 7-8)

Question	Describe your Reasoning
Before the generator is started, state the charge on the belt and each roller?	If the generator is not on then there is no movement, so no charging by friction between the rollers and the belt. Therefore, at the start the belt and rollers will start off as neutral.
Once the generator is started, outline the charge on the bottom silicon roller and the	Since rubber is below silicon on the triboelectric series it has a

belt on the way up?	lower electron affinity. • This means that when rubber and silicon are rubbed together electrons will move from the rubber belt to the Silicon roller making the roller negative and the rubber belt positive as it travels up to the top of the generator
Once the belt encounters the nylon roller and metal brush at the top what is the charge on the belt and nylon roller as the belt goes back down. Outline your reasoning	• Since the rubber belt is positive on the way up to the top of the generator, when it encounters the nylon roller and the metal brush and dome system it sees it as an opportunity to neutralize itself • When in contact with the nylon roller and the metal brush and dome system, both the roller and the brush/dome put negatives on the rubber belt to try to make it neutral • This makes the nylon roller positive and the metal brush and dome positive. • The belt is constantly being recharged therefore it is not able to become neutral, but it does gain a negative charge from all the negatives it is getting from the brush and dome making the belt negative on the way back down to the bottom of the generator
Why does a person's hair stand on end when they touch the dome of this Van De Graff generator? Outline your reasoning	• As described above, The metal dome is positively charged because it has given away electrons to the belt • When a neutral person puts their hands on the ball of the generator, The ball of the generator sees it as a chance to get back to neutral so it takes negatives from the person's hands. • This makes the person positive because they have lost negatives • Even though the dome is taking

	away negatives, It does not gain enough negatives to make itself neutral because it is constantly giving them a way to the belt due to the constant recharging by the machine • Therefore, the dome stays positive and because you are also positive you see a repulsion in your hair because the electrons on you are trying to get as far away from the positive dome as they can
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